

7. The geometry of finite-dimensional Teichmüller spaces

7.1. Finite-Dimensional Teichmüller Spaces

Pro 7.1.1 T_X of a hyperbolic Riemann surface X is finite dimensional iff X is a finite type

If X is of genus g with n points removed,

Then T_X has complex dim $3g-3+n$.

Pf: Finite type: isomorphic to a cpt with a finite number points removed.

Finite type: X is finite type iff it carries a finite geodesic multicurve s.t. all components of the complement are trousers ($S^2 - 3$ points) (3.b.4)

Firstly: If X is not a finite type.

Condition A: X contains an infinite multicurve Γ .

For any $\{\gamma_1, \dots, \gamma_n\} \subset \Gamma$. Their length can change independently $\Rightarrow \{l_1, \dots, l_n\} \Rightarrow T_X$ is infinite dim.

Cor

Condition B: X has non-empty $I(X)$.
 Choose an arbitrary quasi-symmetric homeo
 of $I(X) \simeq \text{id}$.

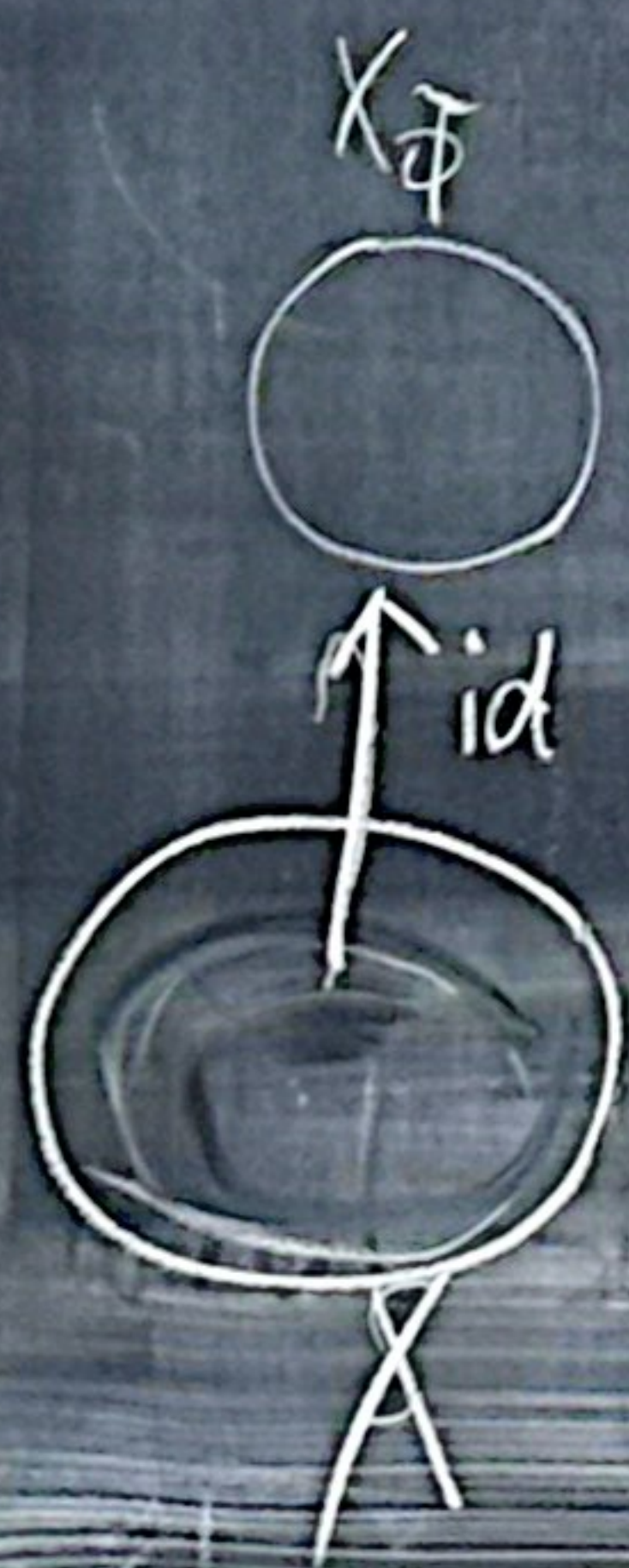
$$\text{skew}(a_1, a_2, a_3) := \sup \frac{|a_i - a_j|}{|a_i - a_k|}$$

$$h: [1, \infty) \rightarrow [1, \infty),$$

$$\text{skew}(f(a_1), f(a_2), f(a_3)) \leq h(\text{skew}(a_1, a_2, a_3))$$

$$\text{skew}(f^{-1}(a'_1), f^{-1}(a'_2), f^{-1}(a'_3)) \leq h(\text{skew}(a'_1, a'_2, a'_3))$$

infinite dim.



Extend it to quasi-conformal homeo
 of X , Since: Space of quasi-symmetric
 map has infinite dim, map:

$$\phi \rightarrow [X_\phi, \text{id}] \text{ in } T_X \text{ has infinite dim}$$

$$[X_\phi, \phi]$$

If $X(g, n)$

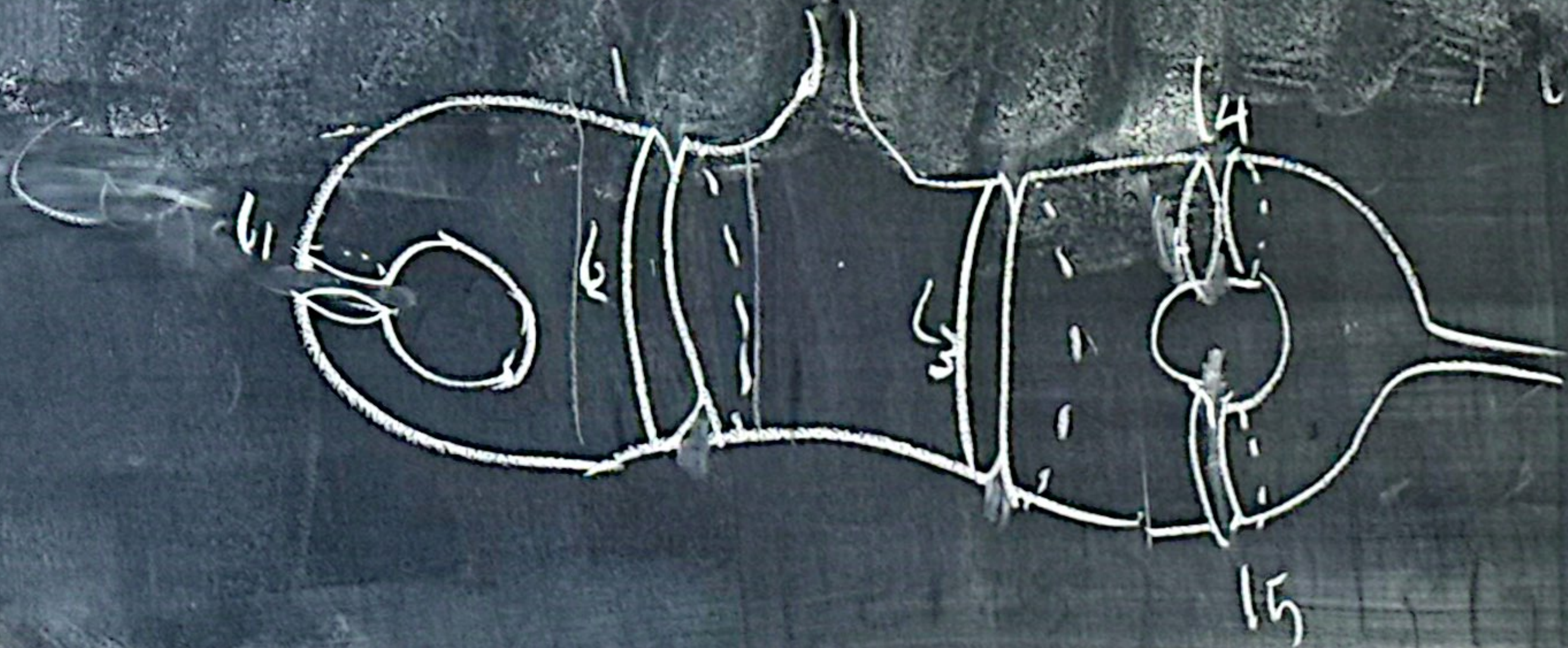
M1: $Q'(X)$ By Riemann-Roch Thm: it has $\dim 3g+n-3$.

$T_x \Leftrightarrow Q'(X) \Rightarrow T_x$ has $\dim 3g+n-3$

M2: For (g, n) -surface. $\exists \Gamma$: multicurve, divide X into trousers. $\Gamma: \gamma_i \hookrightarrow 2$ -parameters. $|\Gamma| = 3g-3+n$



$$\dim_c = \frac{2(3g-3+n)}{2} = 3g-3+n$$



$$g=2 \quad n=2$$
$$|\Gamma|=5$$

$$\frac{2 \times 5}{2} = 5$$

Example 7.1.2 (Homeomorphic Riemann Surfaces (not finite type) belong to different Teichmüller spaces)

Def: $X := \mathbb{C} - (\{2^n | n \in \mathbb{Z}\} \cup \{0\})$ γ_n : circle centered at 0

$$r = \frac{3}{2}$$

$$i) \quad \text{Aut: } z \rightarrow z z^n \quad \gamma_n = 2^n \gamma_0$$

conformal, maps geodesic to geodesic
 $\therefore \{\gamma_n\}$: geodesics

annuli removed a point is a trouser.

For trouser between $\gamma_n, \gamma_{n+1} \rightarrow \mu_n$
 $\Rightarrow$ get $\{\mu_n\}$ as new lengths of γ_n

$$\forall \gamma, \frac{1}{K(\phi)} l_X(\gamma) \leq l_Y(\phi(\gamma)) \leq K(\phi) l_X(\gamma)$$

$$|\ln l_Y(\phi(\gamma)) - \ln l_X(\gamma)| \leq \ln K(\phi)$$

In X : γ_n have same hyp length: L , let $l_Y(\gamma_n) = \ln$

If $\exists$ quasi-conformal map ($\supset id$) $\phi: X \rightarrow Y$

$$\forall n, |\ln \ln - \ln L| \leq \ln K(\phi)$$

If let $\{\ln \ln\}$ be unbounded

$$d((X, id), (Y, \phi)) = \inf_{\phi} \ln(K(\phi)) \quad \text{large } (\infty)$$

$\Rightarrow$ Not in same Teichmüller space

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